

The Next Wave in SciML: Foundation Models for Physics

Understanding Poseidon and the Scalable Operator Transformer (scOT)

Introduction to Scientific Machine Learning

Traditional Simulators: The "Gold Standard"

Finite Element Methods (FEM) and Computational Fluid Dynamics (CFD) solve PDEs from first principles every single time.

Pros:

- Highly accurate, reliable, and trusted by engineers

Cons:

- Incredibly slow (hours to days)
- Computationally expensive
- Bottleneck for rapid iteration

Standard SciML: The "Trained Specialist"

Physics-Informed Neural Networks (PINNs) and Fourier Neural Operators (FNO) learn to solve one specific type of PDE.

Pros:

- Much faster than traditional simulators once trained

Cons:

- Poor generalization—must retrain from scratch for every new problem

Foundation Models: The "Next Generation"

Pre-trained on diverse physics problems, then quickly fine-tuned for specific applications.

Pros:

- Fast inference
- Strong generalization
- Sample efficient

Cons:

- Requires significant upfront computational investment

The Simulation Challenge: Three Ways to Solve Physics

Understanding the trade-offs between accuracy, speed, and generalization is crucial for choosing the right approach to computational physics problems.

Traditional Simulators

The "Gold Standard"

Finite Element Methods (FEM) and Computational Fluid Dynamics (CFD) solve PDEs from first principles every single time.

- **Pros:** Highly accurate, reliable, and trusted by engineers
- **Cons:** Incredibly slow (hours to days), computationally expensive, bottleneck for rapid iteration

A New Approach: The "Foundation Model" for Physics

What is Poseidon?

Poseidon represents a paradigm shift in scientific machine learning—it's a **foundation model for physics**, analogous to how GPT is a foundation model for language. Rather than learning one specific problem, it learns the fundamental "grammar" of physical phenomena.

The "Pre-train Once, Fine-tune Many" Strategy

Pre-train: The "Physics PhD" Model learns general patterns from massive, diverse datasets spanning fluid dynamics, wave propagation, and diffusion processes

Fine-tune: The "New Project" Show the model just a few examples of your specific problem—it adapts rapidly using its foundational knowledge

Deploy: Fast Predictions Get SciML speed without retraining from scratch for each new application

 **Key Insight:** Transfer learning works for physics just like it does for language and vision. The fundamental equations governing natural phenomena share deep mathematical structures.

The Engine: scOT (Scalable Operator Transformer)

Standard Transformers fail at physics because they can't scale to the massive grids (1M+ points) required for 2D/3D simulations. scOT solves this fundamental bottleneck.

1. **The Scaling Problem**Global attention in standard Transformers has $O(N^2)$ complexity—completely intractable for million-point physics grids
2. **Neural Operator Framework**scOT learns mappings between *entire functions* (e.g., complete pressure field at $t=0 \rightarrow$ pressure field at $t=1$), not just point-wise predictions
3. **Windowed Attention**Divides the physics grid into small, manageable windows. Each point only attends to neighbors in its local window—like having stadium sections talk internally rather than everyone shouting across the entire venue
4. **Efficient Computation**This architectural choice breaks the quadratic scaling barrier, enabling efficient processing of high-resolution spatial domains

Poseidon's Key Innovations

Making Foundation Models Smarter and More Data-Efficient

Predicting "In-Between" Time Steps

Problem: Most models only predict at discrete training times ($t=1, 2, 3$) and fail at intermediate points like $t=2.5$

Solution: Time-Conditioned Layer Norms feed the continuous time value directly into the model

Result: Query predictions at *any* continuous time point with high accuracy

Smarter Training Data Re-use

Problem: Physics simulations are expensive to generate, making training data precious and scarce

Solution: "all2all" Training Strategy intelligently slices one long simulation into thousands of overlapping sub-sequences

Result: Extracts maximum learning signal from minimal data—orders of magnitude more efficient

The Payoff: Why This Matters for SciML

Foundation models like Poseidon fundamentally change what's possible in computational physics and engineering design workflows.

20 Training Samples Needed

Poseidon achieves target accuracy with ~20 samples vs. 1,000+ for traditional SciML methods like FNO

100x Speed Improvement

Orders of magnitude faster than traditional solvers—seconds instead of hours for comparable accuracy

95% Accuracy Maintained

Maintains high fidelity predictions while dramatically reducing computational cost

Extreme Sample Efficiency

Pre-training enables few-shot learning. Fine-tune on minimal problem-specific data and achieve production-ready accuracy quickly.

Strong Generalization

The breakthrough advantage: pre-train on general fluid dynamics, then fine-tune for *unseen* problems. Learns transferable physics principles, not memorized solutions.

Incredible Speed

Enables rapid prototyping, real-time design optimization, and interactive "what-if" exploration previously impossible with traditional methods.

Summary & Key Takeaways

1. Traditional Simulators Accurate and trusted, but **too slow** for iterative workflows
 2. Standard SciML Fast inference, but **poor generalization**—must retrain for every problem
 3. Foundation Models **The future**—general, scalable, and efficient
-

The Big Idea

We are transitioning from building *one-off* models for *individual* problems to developing **general foundation models** that understand the universal "language of physics."

- **General** Pre-trained on diverse physics domains for broad applicability
- **Scalable** scOT architecture with windowed attention handles massive 2D/3D grids efficiently
- **Efficient** Minimal data requirements and lightning-fast inference times